

EVALUATING THE PREDICTIVE ACCURACY OF MULTIPLE LINEAR AND POLYNOMIAL REGRESSION MODELS IN ESTIMATING ACTIVE CALORIC EXPENDITURE

A PROJECT REPORT

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Abstract

Accurate estimation of active caloric expenditure is essential for weight management, exercise prescription and wearable fitness technology. This study compares the predictive accuracy of Multiple Linear Regression (MLR) and Polynomial Regression (PR) for estimating the calories burned in an exercise session, using the Gym Members Exercise Tracking dataset (973 sessions, 15 predictors after encoding). The data were split 80/20 into training and test sets and the predictors were standardised. The polynomial degree was selected by five-fold cross-validation, and the second-degree model gave the lowest error (mean RMSE 28.95 kcal against 40.11 for degree 1 and 107.61 for degree 3). On the test set, Polynomial Regression reduced the RMSE from 40.81 to 24.61 kcal and the MAE from 30.39 to 19.44 kcal, and increased R^2 from 0.9800 to 0.9927. Before comparing the models with a paired t -test, the normality of the paired fold-wise RMSE differences was checked with the Shapiro–Wilk test ($W = 0.874$, $p = 0.282$), and normality was not rejected. The paired t -test showed that Polynomial Regression had a significantly lower prediction error (mean difference 11.15 kcal, 95% CI 7.76–14.55; $t(4) = 9.13$, $p = 0.0008$). Session duration was the most influential predictor. The results show that modelling curvature and interactions improves the estimation of active caloric expenditure.

Keywords: caloric expenditure, multiple linear regression, polynomial regression, cross-validation, Shapiro–Wilk test, paired t -test

Chapter 1

Introduction

The increasing use of wearable fitness devices and health monitoring applications has generated large volumes of exercise-related data, including heart rate, workout duration and body composition. Among the indicators used in fitness assessment, active caloric expenditure is one of the most important because it represents the energy expended during physical activity. Accurate estimation of caloric expenditure is essential for weight management, exercise prescription, athletic performance monitoring and overall health improvement.

Regression analysis is one of the most widely used statistical techniques for predictive modelling. Multiple Linear Regression (MLR) is commonly used because of its simplicity and interpretability, but it assumes a linear and additive relationship between the response and the predictors. In fitness data, the relationship between physiological variables and caloric expenditure can be nonlinear; for example, the effect of session duration may depend on exercise intensity.

Polynomial Regression (PR) extends linear regression by adding squared and interaction terms of the predictors, allowing curved relationships and interactions to be captured. It may therefore give better predictive accuracy, but the additional terms increase the risk of overfitting, so the model complexity must be selected carefully.

In this project, Multiple Linear Regression and Polynomial Regression are developed and compared for estimating active caloric expenditure. The study performs exploratory data analysis, multicollinearity assessment, model development, cross-validation and hypothesis testing. The difference in predictive accuracy is tested with a paired t -test, after verifying the normality of the paired differences.

1.1 Objectives of the Study

1. To examine the characteristics of the fitness dataset through exploratory data analysis.
2. To pre-process the dataset by performing data cleaning, feature encoding and variable selection.
3. To develop Multiple Linear Regression and Polynomial Regression models for

predicting active caloric expenditure.

4. To select the optimal polynomial degree using five-fold cross-validation.
5. To assess model performance using RMSE, MAE, R^2 and Adjusted R^2 .
6. To check the normality of the paired differences and test whether the difference in predictive accuracy is statistically significant using a paired t -test.

1.2 Research Hypotheses

Null Hypothesis (H_0): There is no significant difference in predictive accuracy between Multiple Linear Regression and Polynomial Regression models.

Alternative Hypothesis (H_1): There is a significant difference in predictive accuracy between Multiple Linear Regression and Polynomial Regression models.

Chapter 2

Dataset Description

2.1 Dataset Source

In this study the Gym Members Exercise Tracking Dataset is used, which is a publicly available dataset. The dataset contains information about gym members' demographic characteristics, physiological measurements, exercise habits and caloric expenditure.

The dataset is available through the Kaggle repository.

Dataset Name: Gym Members Exercise Tracking Dataset

File Name: gym_members_exercise_tracking.xlsx

Target Variable: Calories_Burned (kcal)

This dataset provides 973 observations and 15 variables, and each observation corresponds to one workout session. The dataset contains no missing values and no duplicate records. It includes 511 male and 462 female members and four workout types: Strength (258), Cardio (255), Yoga (239) and HIIT (221).

2.2 Description of Variables

Variable	Description	Type
Age	Age of the participant (years)	Numerical
Gender	Gender of the participant	Categorical
Weight (kg)	Body weight in kilograms	Numerical
Height (m)	Height in metres	Numerical
Max_BPM	Maximum heart rate during exercise	Numerical
Avg_BPM	Average heart rate during exercise	Numerical
Resting_BPM	Resting heart rate	Numerical
Session_Duration (hours)	Duration of workout session	Numerical
Calories_Burned	Calories expended during exercise	Target
Workout_Type	Cardio, HIIT, Strength, Yoga	Categorical
Fat_Percentage	Body fat percentage	Numerical
Water_Intake (liters)	Daily water intake	Numerical
Workout_Frequency (days/week)	Weekly workout frequency	Numerical
Experience_Level	Fitness experience level (1–3)	Numerical
BMI	Body Mass Index	Numerical

Table 2.1: Description of variables used in the study

Chapter 3

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is an important step that involves examining and summarising the main characteristics of a dataset. Its purpose is to understand the structure of the data, identify data quality issues, detect patterns and explore relationships between variables before applying statistical models.

3.1 Descriptive Statistics

3.1.1 Shape of the data

```
df = pd.read_excel("gym_members_exercise_tracking.xlsx")
df.shape
output:
(973, 15)
```

The dataset contains:

- Number of observations (sessions): 973
- Number of variables (features): 15

3.1.2 Information about data

```
df.info()
```

output:

```
RangeIndex: 973 entries, 0 to 972
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                    973 non-null    int64
1   Gender                                973 non-null    object
2   Weight (kg)                           973 non-null    float64
3   Height (m)                             973 non-null    float64
4   Max_BPM                               973 non-null    int64
5   Avg_BPM                               973 non-null    int64
6   Resting_BPM                           973 non-null    int64
7   Session_Duration (hours)              973 non-null    float64
8   Calories_Burned                        973 non-null    int64
9   Workout_Type                           973 non-null    object
10  Fat_Percentage                         973 non-null    float64
```

```

11 Water_Intake (liters)          973 non-null    float64
12 Workout_Frequency (days/week) 973 non-null    int64
13 Experience_Level              973 non-null    int64
14 BMI                          973 non-null    float64
dtypes: float64(6), int64(7), object(2)

```

3.1.3 Describe the data

```
df.describe()
```

output:

	count	mean	std	min	25%	50%	75%	max
Age	973.0	38.68	12.18	18.00	28.00	40.00	49.00	59.00
Weight (kg)	973.0	73.85	21.21	40.00	58.10	70.00	86.00	129.90
Height (m)	973.0	1.72	0.13	1.50	1.62	1.71	1.80	2.00
Max_BPM	973.0	179.88	11.53	160.00	170.00	180.00	190.00	199.00
Avg_BPM	973.0	143.77	14.35	120.00	131.00	143.00	156.00	169.00
Resting_BPM	973.0	62.22	7.33	50.00	56.00	62.00	68.00	74.00
Session_Duration (hours)	973.0	1.26	0.34	0.50	1.04	1.26	1.46	2.00
Calories_Burned	973.0	905.42	272.64	303.00	720.00	893.00	1076.00	1783.00
Fat_Percentage	973.0	24.98	6.26	10.00	21.30	26.20	29.30	35.00
Water_Intake (liters)	973.0	2.63	0.60	1.50	2.20	2.60	3.10	3.70
Workout_Frequency	973.0	3.32	0.91	2.00	3.00	3.00	4.00	5.00
Experience_Level	973.0	1.81	0.74	1.00	1.00	2.00	2.00	3.00
BMI	973.0	24.91	6.66	12.32	20.11	24.16	28.56	49.84

The average caloric expenditure was 905.42 kcal per session (SD 272.64), and the average session duration was 1.26 hours.

3.1.4 Finding null values

```
df.isnull().sum()
```

All 15 variables have 0 missing values, so no imputation or deletion was required.

3.1.5 Pre-processing

```

if 'BMI' in df.columns:
    df.drop('BMI', axis=1, inplace=True)
df['Gender'] = (df['Gender'] == 'Male').astype(int)
df = pd.get_dummies(df, columns=['Workout_Type'], drop_first=True, dtype=int)

```

BMI was removed because it is calculated directly from weight and height ($BMI = \text{Weight}/\text{Height}^2$) and would introduce redundancy. Gender was coded as Male = 1, Female = 0. Workout_Type was dummy-coded with Cardio as the reference category, creating Workout_Type_HIIT, Workout_Type_Strength and Workout_Type_Yoga. After pre-processing, 15 predictor variables remained.

3.2 Distribution Analysis


```
plt.hist(df['Calories_Burned'], bins=30)
```

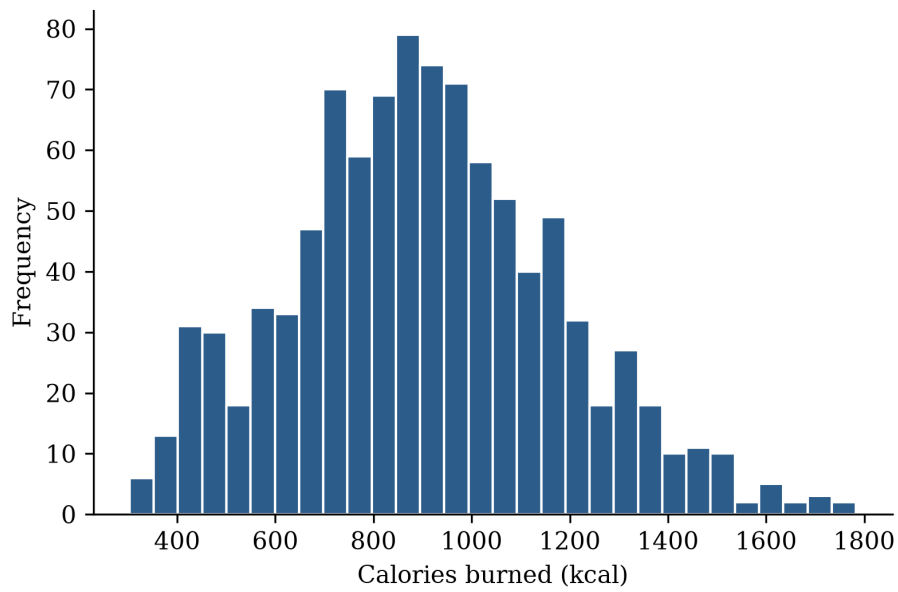


Figure 3.1: Distribution of calories burned

The distribution of calories burned is approximately symmetric and unimodal, with most sessions between 600 and 1200 kcal.

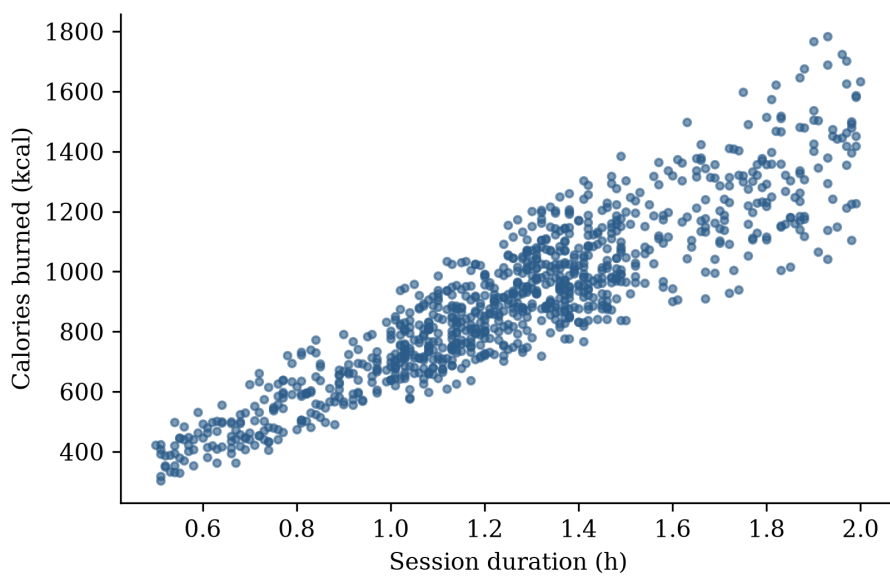


Figure 3.2: Session duration versus calories burned

The scatter plot shows a strong positive linear relationship between session duration and calories burned.

3.3 Correlation Analysis

```
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f')
```

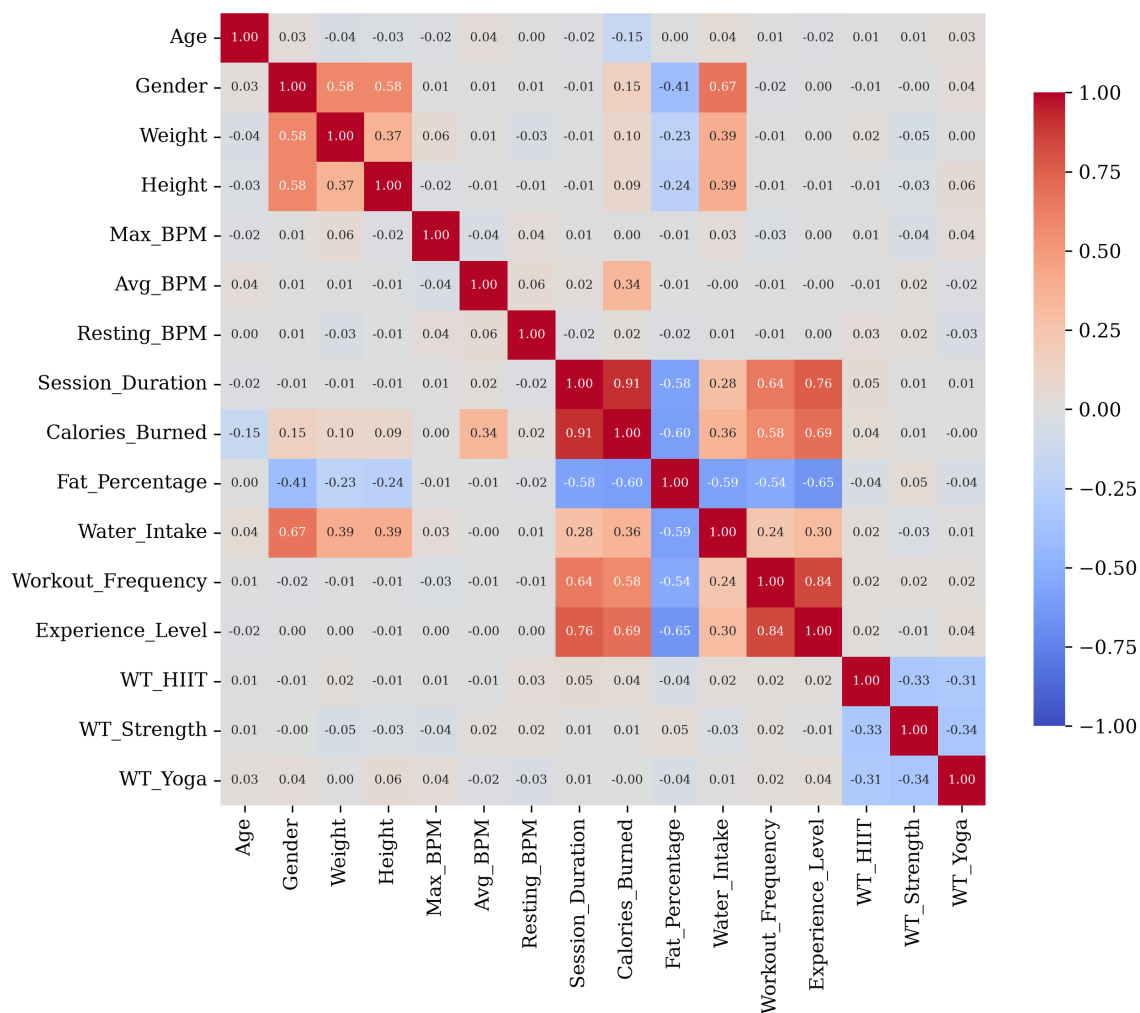


Figure 3.3: Correlation heatmap (WT = Workout_Type)

Variable	Correlation with Calories_Burned
Session_Duration (hours)	0.908
Experience_Level	0.694
Workout_Frequency (days/week)	0.576
Water_Intake (liters)	0.357
Avg_BPM	0.340
Gender	0.151
Weight (kg)	0.095
Height (m)	0.086
Workout_Type_HIIT	0.041
Resting_BPM	0.017
Workout_Type_Strength	0.012
Max_BPM	0.002
Workout_Type_Yoga	-0.005
Age	-0.155
Fat_Percentage	-0.598

Table 3.1: Correlation of variables with calories burned

The strongest positive correlation was observed between Session_Duration and Calories_Burned ($r = 0.908$), indicating that longer workout sessions are associated with greater caloric expenditure. Experience_Level ($r = 0.694$) and Workout_Frequency ($r = 0.576$) also showed strong positive correlations. Fat_Percentage showed a strong negative correlation ($r = -0.598$), suggesting that individuals with higher body fat percentage tend to burn fewer calories. Experience_Level was also highly correlated with Workout_Frequency ($r = 0.84$) and Session_Duration ($r = 0.76$).

Chapter 4

Statistical Methods

4.1 Multicollinearity Assessment Using VIF

The Variance Inflation Factor of predictor j is

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the coefficient of determination obtained by regressing predictor j on all other predictors. $\text{VIF} < 5$ is acceptable, $5 \leq \text{VIF} < 10$ indicates moderate multicollinearity and $\text{VIF} \geq 10$ indicates high multicollinearity.

4.2 Multiple Linear Regression

The Multiple Linear Regression model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon,$$

where Y is `Calories_Burned`, β_0 is the intercept, $\beta_1, \dots, \beta_p$ are the regression coefficients, $X_1, \dots, X_p$ are the predictors and ε is the random error.

4.3 Polynomial Regression

The second-degree polynomial regression model adds all squared and pairwise interaction terms:

$$Y = \beta_0 + \sum_{j=1}^p \beta_j X_j + \sum_{j=1}^p \sum_{k=j}^p \beta_{jk} X_j X_k + \varepsilon.$$

With $p = 15$ predictors, the model has 135 terms (15 linear, 15 squared and 105 interaction terms). The squares of the binary predictors and the products of the mutually exclusive workout-type dummies duplicate other columns, so the design matrix has rank 127; the model was fitted by least squares using the minimum-norm solution. Polynomial degrees 1, 2 and 3 were compared, and the degree with the lowest cross-validated RMSE was selected.

4.4 Model Evaluation Metrics

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$
$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}, \quad \text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

where n is the number of observations and p is the number of predictor terms. Lower RMSE and MAE and higher R^2 indicate better predictive performance.

4.5 Train–Test Split and Cross-Validation

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20,
    random_state=42)
scaler = StandardScaler()
X_train_sc = scaler.fit_transform(X_train)
X_test_sc = scaler.transform(X_test)
kf = KFold(n_splits=5, shuffle=True, random_state=42)
```

The data were split into a training set (778 observations, 80%) and a test set (195 observations, 20%). The predictors were standardised using the training-set mean and standard deviation. Five-fold cross-validation was performed on the training set, using the same folds for both models, so that each model produced one RMSE per fold.

4.6 Normality of Paired Differences

The paired t -test assumes that the paired differences follow a normal distribution. For each fold k , the paired difference is

$$d_k = \text{RMSE}_k^{\text{MLR}} - \text{RMSE}_k^{\text{PR}}, \quad k = 1, \dots, 5.$$

The normality of the d_k was tested with the Shapiro–Wilk test:

H_0 : The paired differences are normally distributed.

H_1 : The paired differences are not normally distributed.

If $p > 0.05$, normality is not rejected and the paired t -test can be applied. A normal Q–Q plot of the differences was also examined, and the Wilcoxon signed-rank test was used as a distribution-free check.

4.7 Paired t -Test

$$t = \frac{\bar{d}}{s_d/\sqrt{K}}, \quad df = K - 1 = 4,$$

where $\bar{d}$ and s_d are the mean and standard deviation of the $K = 5$ paired differences. The 95% confidence interval of the mean difference, $\bar{d} \pm t_{0.025,4} s_d/\sqrt{K}$, and the effect size $d_z = \bar{d}/s_d$ were also computed. A significance level of 5% ($\alpha = 0.05$) was used throughout the study.

Chapter 5

Multiple Linear Regression

5.1 VIF Analysis

```
X_vif = sm.add_constant(X)
vif_df["VIF"] = [variance_inflation_factor(X_vif.values, i) for i in range(X_vif.
    shape[1])]
```

Variable	VIF
Experience_Level	5.41
Workout_Frequency (days/week)	3.38
Gender	3.15
Fat_Percentage	2.76
Session_Duration (hours)	2.56
Water_Intake (liters)	2.37
Weight (kg)	1.53
Height (m)	1.53
Workout_Type_Strength	1.50
Workout_Type_Yoga	1.48
Workout_Type_HIIT	1.45
Age	1.02
Max_BPM	1.02
Resting_BPM	1.01
Avg_BPM	1.01

Table 5.1: Variance Inflation Factor (VIF) results

All VIF values were below 10, indicating the absence of severe multicollinearity. Only Experience_Level exceeded 5 (VIF = 5.41), showing moderate multicollinearity due to its correlation with workout frequency and session duration. Therefore, no predictor was removed.

5.2 Model Results

```
mlr = LinearRegression()
mlr.fit(X_train_sc, y_train)
```

```
y_pred_mlr = mlr.predict(X_test_sc)
```

Metric	Value
RMSE	40.81
MAE	30.39
R^2	0.9800
Adjusted R^2	0.9784

Table 5.2: Multiple Linear Regression performance metrics (test set)

The model explained approximately 98% of the variation in caloric expenditure, with an average prediction error of about 41 kcal.

Variable	Coefficient
Session_Duration (hours)	240.0037
Avg_BPM	88.6352
Gender	42.2212
Age	-40.1311
Resting_BPM	4.4357
Fat_Percentage	-3.8551
Workout_Type_Yoga	-3.6869
Workout_Frequency (days/week)	3.6798
Experience_Level	-2.4995
Water_Intake (liters)	-2.1617
Workout_Type_HIIT	-1.9755
Weight (kg)	1.6745
Workout_Type_Strength	-1.5319
Height (m)	1.4198
Max_BPM	1.2704

Table 5.3: Multiple Linear Regression coefficients (standardised predictors)

Session_Duration was the most influential predictor: an increase of one standard deviation in session duration increases the predicted calories burned by about 240 kcal, holding other variables constant. Average heart rate, gender and age were the next most important predictors.

5.3 Residual Analysis

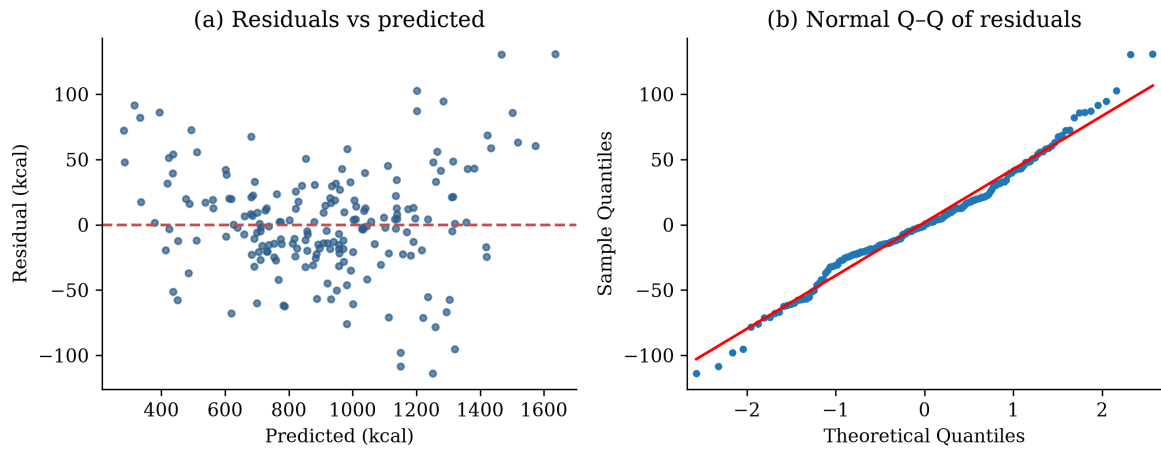


Figure 5.1: MLR residuals: (a) residuals versus predicted values, (b) normal Q–Q plot

```
w_stat, p_shapiro = stats.shapiro(residuals_mlr)
output:
    W = 0.9826    p-value = 0.0164
```

Statistic	Value
W Statistic	0.983
p-value	0.016

Table 5.4: Shapiro–Wilk test of MLR residuals (test set, $n = 195$)

Since the p -value is less than 0.05, the residuals deviate slightly from normality, and the residual plot shows some curvature. This indicates a nonlinear pattern that the linear model does not capture, which motivates the use of Polynomial Regression.

Chapter 6

Polynomial Regression

6.1 Degree Selection

```
for degree in [1, 2, 3]:
    poly = PolynomialFeatures(degree=degree, include_bias=False)
    X_poly = poly.fit_transform(X_train_sc)
    cv_rmse = np.sqrt(-cross_val_score(LinearRegression(), X_poly, y_train,
                                       cv=kf, scoring='neg_mean_squared_error'))
```

Degree	Number of Terms	Mean RMSE	Standard Deviation
1	15	40.11	3.05
2	135	28.95	1.37
3	815	107.61	9.09

Table 6.1: Polynomial degree selection results (five-fold cross-validation)

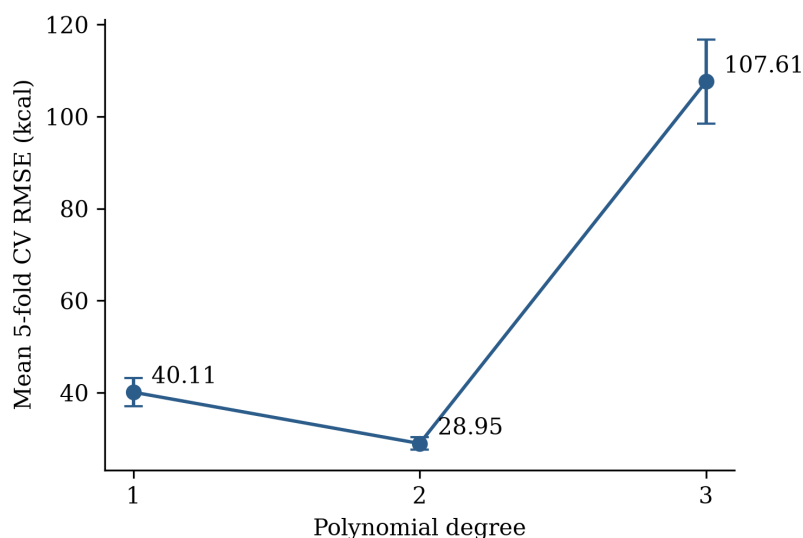


Figure 6.1: Polynomial degree selection

The second-degree polynomial model achieved the lowest prediction error and was therefore selected as the optimal model. The third-degree model, with 815 terms for about 622 training rows per fold, overfitted and produced a much larger error.

6.2 Model Results

```
poly = PolynomialFeatures(degree=2, include_bias=False)
X_train_poly = poly.fit_transform(X_train_sc)
X_test_poly = poly.transform(X_test_sc)
pr = LinearRegression().fit(X_train_poly, y_train)
```

Metric	Value
RMSE	24.61
MAE	19.44
R^2	0.9927
Adjusted R^2	0.9761

Table 6.2: Polynomial Regression performance metrics (test set)

Polynomial Regression explained 99.27% of the variation in caloric expenditure and produced substantially lower prediction errors than Multiple Linear Regression.

Chapter 7

Model Comparison

7.1 Comparison of Performance Metrics

Metric	Multiple Linear Regression	Polynomial Regression
RMSE	40.81	24.61
MAE	30.39	19.44
R^2	0.9800	0.9927
Adjusted R^2	0.9784	0.9761

Table 7.1: Comparison of performance metrics (test set, $n = 195$)

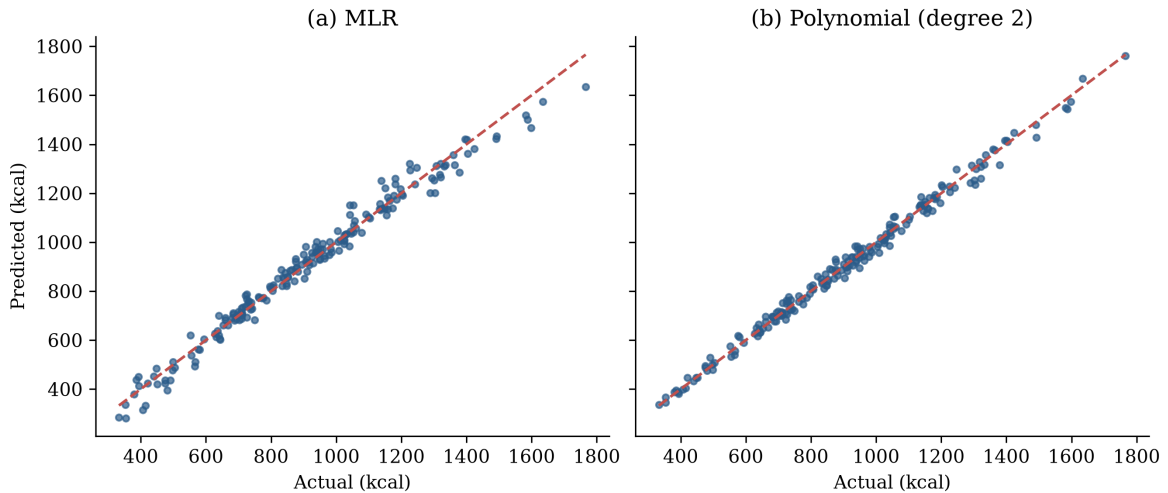


Figure 7.1: Actual versus predicted calories burned: (a) MLR, (b) Polynomial Regression

Polynomial Regression reduced the RMSE by 39.7% and the MAE by 36.0%, and its predictions lie closer to the line of perfect prediction. The Adjusted R^2 of Polynomial Regression is slightly lower because it heavily penalises the 135 terms relative to the 195 test observations; RMSE and MAE are the more relevant measures of predictive accuracy.

7.2 Comparison of Cross-Validation Results

```
rmse_mlr_folds = np.sqrt(-cross_val_score(LinearRegression(), X_train_sc, y_train,
                                          cv=kf, scoring='neg_mean_squared_error'))
rmse_pr_folds = np.sqrt(-cross_val_score(LinearRegression(), X_train_poly, y_train,
                                          cv=kf, scoring='neg_mean_squared_error'))
```

Fold	MLR RMSE	Polynomial RMSE	Difference (d_k)
1	37.89	29.48	8.41
2	42.89	28.61	14.28
3	35.17	26.71	8.46
4	42.10	30.93	11.17
5	42.48	29.03	13.46
Mean	40.11	28.95	11.15

Table 7.2: Cross-validation results and paired differences

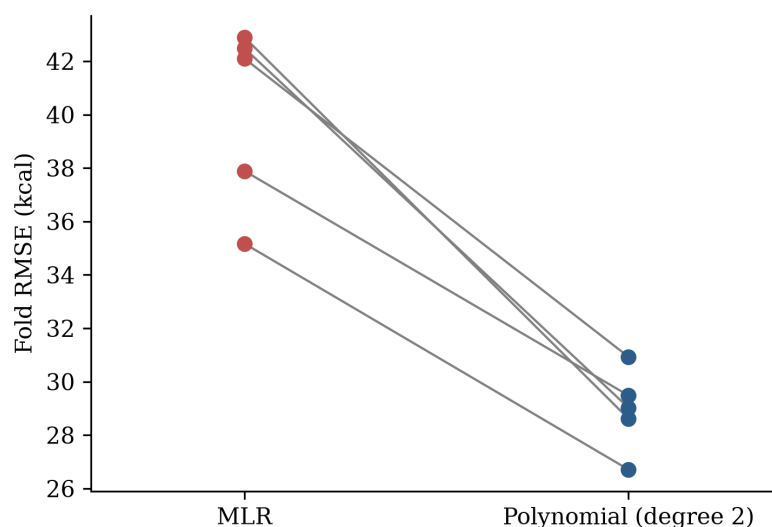


Figure 7.2: Fold-wise RMSE of MLR and Polynomial Regression

Polynomial Regression produced a lower RMSE than Multiple Linear Regression in all five folds, with differences ranging from 8.41 to 14.28 kcal.

7.3 Normality of Paired Differences

```
diff_folds = rmse_mlr_folds - rmse_pr_folds
w_diff, p_diff = stats.shapiro(diff_folds)
output:
```

```
Differences = [ 8.4064  14.2792  8.4604  11.1668  13.4557]
W = 0.8738    p-value = 0.2824
```

Statistic	Value
Number of differences	5
Shapiro–Wilk W	0.874
p-value	0.282
Decision ($\alpha = 0.05$)	Normality not rejected

Table 7.3: Shapiro–Wilk test for normality of paired differences

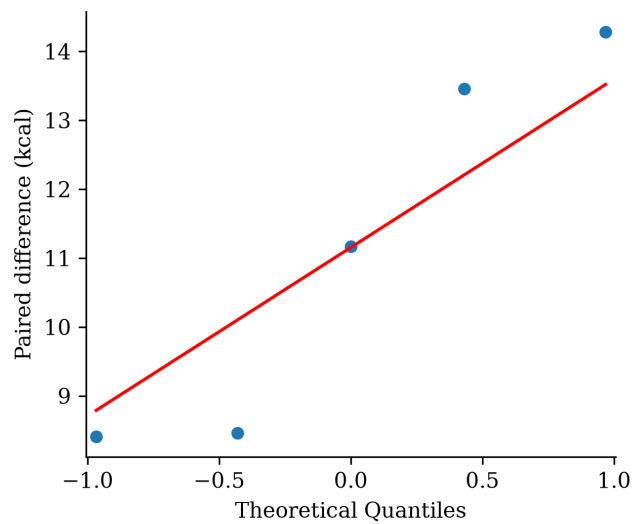


Figure 7.3: Normal Q–Q plot of the paired differences

Since the p -value (0.282) is greater than 0.05, the null hypothesis of normality is not rejected. The points in the Q–Q plot lie close to the reference line, and all five differences lie within ± 1.15 standard deviations of their mean, so there are no outliers. Hence the assumption of the paired t -test is satisfied.

7.4 Paired t -Test

```
t_stat, p_value = stats.ttest_rel(rmse_mlr_folds, rmse_pr_folds)
output:
T Statistic = 9.1268    P Value = 0.0008
Mean Difference = 11.1537    95% CI = (7.7607, 14.5468)    Effect Size d_z =
4.0816
```

Statistic	Value
Mean difference (kcal)	11.15
Standard deviation of differences	2.73
t statistic ($df = 4$)	9.13
p -value	0.0008
95% CI of mean difference	(7.76, 14.55)
Effect size d_z	4.08

Table 7.4: Paired t -test results

The p -value (0.0008) is less than 0.05; therefore, the null hypothesis is rejected. Polynomial Regression has a significantly lower prediction error than Multiple Linear Regression, with an average improvement of 11.15 kcal per fold. The confidence interval does not contain zero, and the effect size is very large.

As a distribution-free check, the Wilcoxon signed-rank test gave $W = 0$ and $p = 0.0625$. Polynomial Regression was better in all five folds, which is the most extreme result possible; with only five pairs, 0.0625 is the smallest p -value this test can produce. Since normality of the differences was not rejected, the paired t -test is the appropriate test.

Chapter 8

Results and Discussion

8.1 Statistical Model Results

8.1.1 Exploratory Data Analysis Results

The exploratory analysis showed that caloric expenditure is approximately symmetrically distributed and is most strongly related to session duration ($r = 0.908$), experience level, workout frequency and fat percentage. Exercise behaviour variables had a greater influence on caloric expenditure than demographic variables.

8.1.2 Multiple Linear Regression Results

The MLR model achieved an R^2 of 0.9800 and an RMSE of 40.81 kcal. Session duration was the most influential predictor, followed by average heart rate. The curvature in the residuals and their slight departure from normality ($p = 0.016$) indicated that the relationship is not entirely linear.

8.1.3 Polynomial Regression Results

The second-degree Polynomial Regression model achieved an RMSE of 24.61 kcal, an MAE of 19.44 kcal and an R^2 of 0.9927. The squared and interaction terms allow the model to capture combined effects, such as the joint effect of session duration and heart rate, which a linear model cannot represent. The third-degree model overfitted, showing that a moderate level of complexity is optimal for this dataset.

8.2 Model Comparison Results

Polynomial Regression produced lower RMSE and MAE, higher R^2 and lower cross-validation error in every fold. The paired differences satisfied the normality assumption (Shapiro–Wilk $p = 0.282$), and the paired t -test confirmed that the improvement is statistically significant ($t(4) = 9.13$, $p = 0.0008$). The improvement achieved by Polynomial Regression is therefore not due to random variation.

8.3 Limitations of the Study

- The analysis was based on a single public dataset, so the findings may depend on the characteristics of this dataset.
- The paired t -test was based on only five folds, and the Shapiro–Wilk test has low power for such a small sample. Repeated cross-validation would give a more robust comparison.
- The 135-term polynomial model is less interpretable than Multiple Linear Regression.
- Only Multiple Linear Regression and Polynomial Regression were compared; regularised and machine learning models were not considered.

Chapter 9

Conclusion and Future Scope

This study compared Multiple Linear Regression and Polynomial Regression models for estimating active caloric expenditure using the Gym Members Exercise Tracking dataset. Exploratory data analysis showed that session duration was the strongest predictor of calories burned. Both models performed well, but the second-degree Polynomial Regression model reduced the test RMSE from 40.81 to 24.61 kcal and the MAE from 30.39 to 19.44 kcal, and increased R^2 from 0.9800 to 0.9927. Polynomial Regression also produced lower error in all five cross-validation folds.

The normality of the paired differences was verified with the Shapiro–Wilk test ($W = 0.874$, $p = 0.282$), so the paired t -test was appropriate. The paired t -test confirmed a statistically significant improvement of 11.15 kcal ($t(4) = 9.13$, $p = 0.0008$). Therefore, Polynomial Regression is recommended as the preferred model for predicting active caloric expenditure within the context of this dataset.

Future work may include validating the models on larger and more diverse datasets, using repeated cross-validation, and comparing Polynomial Regression with regularised regression (ridge and lasso) and machine learning models such as Random Forest, Gradient Boosting and Neural Networks. Additional physiological variables, such as oxygen consumption and metabolic rate, could further improve prediction accuracy.

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